

AMATH 516: Homework 1
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1. **Exercise 1.1:** Given a collection of real $m \times n$ matrices $A_1, A_2, \dots, A_l$, define the linear mapping $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^l$ by setting

$$\mathcal{A}(X) = (\langle A_1, X \rangle, \langle A_2, X \rangle, \dots, \langle A_l, X \rangle)$$

Show that the adjoint is the mapping $\mathcal{A}^*y = y_1A_1 + y_2A_2 + \dots + y_lA_l$.

Recall the definition of an adjoint: For any linear mapping $\mathcal{A} : \mathbb{E} \rightarrow \mathbb{Y}$, there exists a unique linear mapping $\mathcal{A}^* : \mathbb{Y} \rightarrow \mathbb{E}$, called the adjoint, satisfying:

$$\langle \mathcal{A}x, y \rangle = \langle x, \mathcal{A}^*y \rangle \quad \text{for all points } x \in \mathbb{E}, y \in \mathbb{Y}$$

Proof. Let $X \in \mathbb{R}^{m \times n}$ and $y \in \mathbb{R}^l$. Then, it follows:

$$\begin{aligned} \langle \mathcal{A}(X), y \rangle &= \langle (\langle A_1, X \rangle, \langle A_2, X \rangle, \dots, \langle A_l, X \rangle), y \rangle \\ &= \sum_{i=1}^l \langle A_i, X \rangle \cdot y_i \\ &= \left\langle \sum_{i=1}^l y_i A_i, X \right\rangle \\ &= \langle \mathcal{A}^*y, X \rangle \end{aligned}$$

We see that $\langle \mathcal{A}(X), y \rangle = \langle \mathcal{A}^*y, X \rangle$ for all $X \in \mathbb{R}^{m \times n}$ and for all $y \in \mathbb{R}^l$. Hence, by definition of the adjoint, $\mathcal{A}^*(y) = \sum_{i=1}^l y_i A_i$, or in other words, $\mathcal{A}^*y = y_1A_1 + y_2A_2 + \dots + y_lA_l$. $\square$

Extra Credit: After completing the proof, I asked ChatGPT to do the question and this is what it stated:

Given a collection of real $m \times n$ matrices $A_1, A_2, \dots, A_l$, define the linear mapping $\mathcal{A} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^l$ by

$$\mathcal{A}(X) = (\langle A_1, X \rangle, \langle A_2, X \rangle, \dots, \langle A_l, X \rangle),$$

where $\langle A_i, X \rangle = \text{trace}(A_i X^T)$ is the Frobenius inner product.

\begin{proof} Let $X \in \mathbb{R}^{m \times n}$, and let $y \in \mathbb{R}^l$. Then: $\langle \mathcal{A}(X), y \rangle = \sum_{i=1}^l y_i \langle A_i, X \rangle = \sum_{i=1}^l y_i \text{trace}(A_i X^T) = \text{trace}(\sum_{i=1}^l y_i A_i X^T) = \text{trace}(X^T \sum_{i=1}^l y_i A_i) = \langle X, \sum_{i=1}^l y_i A_i \rangle$. Thus, by the definition of the adjoint, we have:

$$\mathcal{A}^*(y) = \sum_{i=1}^l y_i A_i^T.$$

\end{proof}

Here, the error is that $\mathcal{A}^*(y) = \sum_{i=1}^l y_i A_i$ and contrary to what ChatGPT said, $\mathcal{A}^*(y) \neq \sum_{i=1}^l y_i A_i^T$.

2. **Exercise 1.2:** Given a positive definite matrix $A \in \mathbb{S}^n$, show that the assignment $\langle v, w \rangle_A = \langle Av, w \rangle$ is an inner product on $\mathbb{R}^n$ with the induced norm $\|v\|_A = \sqrt{\langle Av, v \rangle}$. Show that the dual norm with respect to the original inner product $\langle \cdot, \cdot \rangle$ is $\|v\|_A^* = \|v\|_{A^{-1}} = \sqrt{\langle A^{-1}v, v \rangle}$. **Hint:** Use the fact that any positive definite matrix A admits a square root.

From the definition of inner product, the inner product satisfies the following properties:

1. Symmetric Property
2. Bilinearity
3. Positive Definite Property: For any $u \in V$, $\langle u, u \rangle \geq 0$ and $\langle u, u \rangle = 0$ if and only if $u = 0$

We will prove that the properties desired are satisfied:

1. Symmetric Property: Since A is self adjoint and the original inner product is symmetric, we have:

$$\langle v, w \rangle_A = \langle Av, w \rangle = \langle v, A^*w \rangle = \langle v, Aw \rangle = \langle Aw, v \rangle = \langle w, v \rangle_A$$

Thus, symmetry is satisfied.

2. Bilinearity

We have that $\langle av_1 + bv_2, w \rangle_A = \langle A(av_1 + bv_2), w \rangle$ as $\langle \cdot, \cdot \rangle_A$ is defined as $\langle v, w \rangle_A = \langle Av, w \rangle$. Now, $\langle A(av_1 + bv_2), w \rangle = \langle aAv_1 + bAv_2, w \rangle$. We also can express this as $\langle aAv_1 + bAv_2, w \rangle = a\langle Av_1, w \rangle + b\langle Av_2, w \rangle$. Now, this term $a\langle Av_1, w \rangle + b\langle Av_2, w \rangle = a\langle v_1, w \rangle_A + b\langle v_2, w \rangle_A$ as we apply the definition of the inner product and linear map. Thus, we have that $\langle av_1 + bv_2, w \rangle_A = a\langle v_1, w \rangle_A + b\langle v_2, w \rangle_A$ proving that we have bilinearity as desired.

3. Positive Definite Property: For any $u \in V$, $\langle u, u \rangle \geq 0$ and $\langle u, u \rangle = 0$ if and only if $u = 0$

Since A is positive definite for all $v \neq 0$, we have that $\langle v, v \rangle_A = \langle Av, v \rangle$ and we have that $\langle Av, v \rangle > 0$ as it is given that A is a positive definite matrix, which means that for all nonzero $v \in \mathbb{R}^n$, the quadratic form $v^T Av = \langle Av, v \rangle$ is actually positive. Thus, we have that the Positive Definite Property is satisfied.

Now, we show that the dual norm with respect to the original inner product $\langle \cdot, \cdot \rangle$ is $\|v\|_A^* = \|v\|_{A^{-1}} = \sqrt{\langle A^{-1}v, v \rangle}$. Remember that A is defined as a positive definite matrix such that $A \in \mathbb{S}^n$. We will define a matrix B such that it is the symmetric positive definite square root of A meaning that $A = B^T B$ and $A = B^2$. Now, we let $y = Bx$ and this means that $x = B^{-1}y$. Then, it follows that:

$$\langle v, x \rangle = \langle v, B^{-1}y \rangle = \langle B^{-T}v, y \rangle$$

Additionally, we have that:

$$\begin{aligned} \langle Ax, x \rangle &= x^T Ax \\ &= x^T (B^T B)x && (\text{as } A = B^T B) \\ &= x^T B^T Bx && (\text{Associative Property}) \\ &= (Bx)^T Bx && (x^T B^T = (Bx)^T) \\ &= \|Bx\|^2 \\ &= \|y\|^2 \end{aligned}$$

So, the constraint becomes $\|y\| \leq 1$, so it follows:

$$\|v\|_A^* = \sup_{\|y\| \leq 1} \langle B^{-T}v, y \rangle = \|B^{-T}v\| = \sqrt{\langle B^{-T}v, B^{-T}v \rangle}$$

Note that $\sqrt{\langle B^{-T}v, B^{-T}v \rangle}$ is another way of stating $\sqrt{\langle A^{-1}v, v \rangle}$ because $\langle B^{-T}v, B^{-T}v \rangle = \langle A^{-1}v, v \rangle$.

Finally, since $B^{-T}v = (B^{-1})^T v$, and $A^{-1} = (B^{-1})^T (B^{-1})$, we have:

$$\|B^{-T}v\|^2 = \langle B^{-T}v, B^{-T}v \rangle = \langle A^{-1}v, v \rangle$$

Thus,

$$\|v\|_A^* = \sqrt{\langle A^{-1}v, v \rangle} = \|v\|_{A^{-1}}$$

We have shown that the dual norm with respect to the original inner product $\langle \cdot, \cdot \rangle$ is $\|v\|_A^* = \|v\|_{A^{-1}} = \sqrt{\langle A^{-1}v, v \rangle}$.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

✅ **Step 4: Use Change of Variables with Square Root of A (Hint)**

Since $A \succ 0$, there exists a unique symmetric positive definite square root $A^{1/2}$ such that $A = A^{1/2} A^{1/2}$. Let's use the change of variables:

$$\text{Let } y = A^{1/2}x \Rightarrow x = A^{-1/2}y$$

Now observe:

$$\bullet \quad \|x\|_A = \sqrt{x^T A x} = \sqrt{x^T A^{1/2} A^{1/2} x} = \|A^{1/2}x\|_2 = \|y\|_2$$

So the unit sphere $\mathbb{S}_A^n$ becomes $\{x = A^{-1/2}y : \|y\|_2 = 1\}$

Let's write the dual norm as:

$$\|v\|_A^* = \sup_{\|x\|_A=1} \langle v, x \rangle = \sup_{\|y\|_2=1} \langle v, A^{-1/2}y \rangle = \sup_{\|y\|_2=1} \langle A^{-1/2}v, y \rangle$$

This is the **standard Euclidean dual norm** $\|A^{-1/2}v\|_2$, since for any vector z , we have:

$$\sup_{\|y\|_2=1} \langle z, y \rangle = \|z\|_2$$

So finally:

$$\|v\|_A^* = \|A^{-1/2}v\|_2 = \sqrt{v^T A^{-1}v} = \|v\|_{A^{-1}}$$

The problem here is that for our purposes $\mathbb{S}^n$ is defined as the set of $n \times n$ real symmetric matrices rather than how ChatGPT interpreted it as meaning the unit sphere, which leads to a mistake in doing the exercise.

3. **Exercise 1.5:** Show that a function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is closed if and only if the set, $\{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$, is closed.

We will do this Proof by Biconditional:

Showing that if a function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is closed, then the set, $\{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$, is closed:

Proof. Suppose $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is closed. By definition, this means that the epigraph of f , defined as $\text{epi}(f) = \{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$, is a closed subset of $\mathbb{E} \times \mathbb{R}$. Therefore, the set $\{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$ is closed, as desired. $\square$

Showing that if the set, $\{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$, is closed, then our function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is closed:

Proof. Suppose the set $\mathcal{S} = \{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$ is closed. We want to show that f is closed in the sense that its sublevel sets denoted by $L^-(f, r)$ and defined as $L^-(f, r) = \{x \in \mathbb{E} : f(x) \leq r\}$ are closed under limits for every $r \in \mathbb{R}$. Let $(x_k) \subseteq \mathbb{E}$ be a sequence such that $x_k \rightarrow x$ and $f(x_k) \rightarrow \ell \in \mathbb{R}$. We will show that $f(x) \leq \ell$. Since $f(x_k) \rightarrow \ell$, we have $f(x_k) \leq r_k$ and $r_k = \ell + \frac{1}{k}$ for all sufficiently large k . So for large k , $(x_k, r_k) \in \mathcal{S}$, and clearly $(x_k, r_k) \rightarrow (x, \ell)$. Since $\mathcal{S}$ is closed, this limit point $(x, \ell) \in \mathcal{S}$. Therefore, $f(x) \leq \ell$, where $\ell = \lim f(x_k)$, which shows that f is lower semicontinuous, and hence closed. $\square$

Thus, by showing that if a function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is closed, then the set, $\{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$, is closed and showing that if the set, $\{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$, is closed, then our function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is closed, we have shown that a function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is closed if and only if the set, $\{(x, r) \in \mathbb{E} \times \mathbb{R} : f(x) \leq r\}$, is closed.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

A function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is closed if and only if the fibered quantum field of allowed energies over space $\mathbb{E}$, defined by its epigraph, yields a topologically closed space of physical states. That is, no path in state space exits the system — the boundary of accessible configurations lies within the system itself. This quantum stability corresponds precisely to the closedness of the epigraph, and hence of the function f .

The problem here is that ChatGPT is attempting to use a fibered quantum field in order to do the problem, which is not only irrelevant, but also erroneous; therefore, this is a mistake; rather, we should start by using the definitions in general topology, such as open and closed sets, sublevel sets, epigraphs, and limits.

4. **Exercise 1.6:** Existence of minimizers on compact sets: Consider a closed function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ and a nonempty compact set $Q \subset \mathbb{E}$. Then the infimum value $\inf_{x \in Q} f(x)$ is attained at some point in Q . Hint: Apply the Bolzano-Weierstrass Theorem to the sequence $x_i \in Q$ satisfying $f(x_i) \rightarrow \inf_Q f$ and invoke lower semicontinuity.

We first construct a sequence in order to make it clear what we are working with. So, we define a sequence $\{x_i\}_{i=1}^\infty$ in Q such that $f(x_i) \rightarrow \inf_{x \in Q} f(x)$ as $i \rightarrow \infty$. Since Q is compact, the sequence $\{x_i\}$ has a convergent subsequence $\{x_{i_k}\}$. By the Bolzano-Weierstrass Theorem, the subsequence $\{x_{i_k}\}$ converges to a point $x^* \in Q$ as Q is compact. In other words, because Q is compact, by the Bolzano-Weierstrass Theorem, there exists a subsequence $\{x_{i_k}\}$ and a point $x^* \in Q$ such that $\{x_{i_k}\} \rightarrow x^*$. Now, since f is closed, or lower semicontinuous, we have that:

$$\begin{aligned} f(x^*) &\leq \liminf_{k \rightarrow \infty} f(\{x_{i_k}\}) \\ &= \lim_{i \rightarrow \infty} f(x_i) \end{aligned}$$

By the definition of infimum, $f(x^*) \geq \lim_{i \rightarrow \infty} f(x_i)$. Therefore, by the Trichotomy Property, because $f(x^*) \geq \lim_{i \rightarrow \infty} f(x_i)$ and $f(x^*) \leq \lim_{i \rightarrow \infty} f(x_i)$, it must be the fact that $f(x^*) = \lim_{i \rightarrow \infty} f(x_i)$ thus showcasing that x^* is our minimizer for our function f . Therefore, the infimum of f on Q is attained at $x^* \in Q$.

Extra Credit: After completing the exercise, I asked ChatSonic to do the question and this is what it stated:

Existence of minimizers on compact space: Consider a closed function $f: \mathbb{E} \rightarrow \overline{\mathbb{R}}$ and a nonempty compact set $Q \subset \mathbb{E}$. Then the infimum value $\inf_{x \in Q} f(x)$ is attained at some point in Q . Hint: Apply the Bolzano-Weierstrass Theorem to the sequence $x_i \in Q$ satisfying $f(x_i) \rightarrow \inf_Q f$ and invoke lower semicontinuity.

I'll provide a proof for the existence of minimizers on compact spaces.

Theorem: Consider a closed (lower semicontinuous) function $f: \mathbb{E} \rightarrow \overline{\mathbb{R}}$ and a nonempty compact set $Q \subset \mathbb{E}$. Then the infimum value $\inf_{x \in Q} f(x)$ is attained at some point in Q .

Proof:

1. Let $\alpha = \inf_{x \in Q} f(x)$. Since Q is nonempty, α is well-defined (though possibly $-\infty$).
2. Choose a sequence $\{x_i\}_{i=1}^{\infty} \subset Q$ such that $f(x_i) \rightarrow \alpha$ as $i \rightarrow \infty$. This is possible by the definition of infimum.
3. Since Q is compact and $\{x_i\} \subset Q$, by the Bolzano-Weierstrass Theorem, there exists a subsequence $\{x_{i_k}\} \subset \{x_i\}$ that converges to some point $x^* \in Q$.

Proof:

1. Let $\alpha = \inf_{x \in Q} f(x)$. Since Q is nonempty, α is well-defined (though possibly $-\infty$).
2. Choose a sequence $\{x_i\}_{i=1}^{\infty} \subset Q$ such that $f(x_i) \rightarrow \alpha$ as $i \rightarrow \infty$. This is possible by the definition of infimum.
3. Since Q is compact and $\{x_i\} \subset Q$, by the Bolzano-Weierstrass Theorem, there exists a subsequence $\{x_{i_k}\}$ of $\{x_i\}$ that converges to some point $x^* \in Q$.
4. Since f is lower semicontinuous, we have:
$$f(x^*) \leq \liminf_{k \rightarrow \infty} f(x_{i_k})$$
5. But $f(x_{i_k}) \rightarrow \alpha$ as $k \rightarrow \infty$ (since this is a subsequence of the original sequence), so:
$$f(x^*) \leq \lim_{k \rightarrow \infty} f(x_{i_k}) = \alpha$$
6. However, since $x^* \in Q$ and $\alpha = \inf_{x \in Q} f(x)$, we also have $f(x^*) \geq \alpha$.
7. Combining steps 5 and 6, we get $f(x^*) = \alpha$.
8. Therefore, the infimum value α is attained at the point $x^* \in Q$.

This proves that for a lower semicontinuous function on a compact set, the infimum is always attained.



The problem here is the second bullet point:

Choose a sequence $\{x_i\}_{i=1}^{\infty} \subset Q$ such that $f(x_i) \rightarrow \alpha$ as $i \rightarrow \infty$.
This is possible by the definition of infimum.

The definition of infimum only says:

$$\forall \epsilon > 0, \exists x \in Q \text{ such that } f(x) < \alpha + \epsilon$$

The problem is that this does not guarantee the existence of a sequence $x_i \in Q$ such that $f(x_i) \rightarrow \alpha$ unless $f(x) > \alpha$ is somewhere nearby, or more mathematically speaking, in the topological neighborhood; this is the mistake. Instead, to construct a sequence, we must explicitly use the definition of infimum repeatedly for a sequence $\epsilon_i \rightarrow 0$. More rigorously, we must have that for each $i \in \mathbb{N}$, choose $x_i \in Q$ such that $f(x_i) < \alpha + \frac{1}{i}$. This ensures that $f(x_i) \rightarrow \alpha$ as $i \rightarrow \infty$; this rigor gives a better landscape with respect to the infimum we are working with and moreover, is necessary in order to construct our sequence. Thus, we should say:

By the definition of infimum, for each $i \in \mathbb{N}$,
there exists $x_i \in Q$ such that $f(x_i) < \alpha + \frac{1}{i}$.
Thus, the sequence $\{x_i\} \subset Q$ satisfies $f(x_i) \rightarrow \alpha$.

5. **Exercise 1.8:** Existence of unconstrained minimizers: Any coercive closed function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ has a minimizer. **Hint:** Choose $r \in \mathbb{R}$ such that the sublevel set $\mathcal{L} = \{x : f(x) \leq r\}$ is nonempty and apply the existence of minimizers on compact sets.

Choose $r \in \mathbb{R}$ such that the sublevel set $\mathcal{L} = \{x : f(x) \leq r\}$ is nonempty. Since f is coercive, we can say that the sublevel set $\mathcal{L} = \{x : f(x) \leq r\}$ is bounded for every $r \in \mathbb{R}$. We will show that our sublevel set $\mathcal{L}$ is closed.

Proof. A function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is lower semicontinuous at a point $x \in \mathbb{E}$ if we have that $\liminf_{y \rightarrow x} f(y) \geq f(x)$. Equivalently, for every $r \in \mathbb{R}$, the set $\{x \in \mathbb{E} : f(x) > r\}$ is open. Taking complements, $\{x \in \mathbb{E} : f(x) \leq r\}$ is closed. So, we have that our sublevel set of f is closed. $\square$

Because we have $\mathcal{L}$ as both bounded and closed, we have that $\mathcal{L}$ is compact by the Heine-Borel Theorem. Applying the result from **Exercise 1.6**, we conclude that f attains its minimum on $\mathcal{L}$. Furthermore, there exists $x^* \in \mathcal{L}$ such that $f(x^*) = \min_{x \in \mathcal{L}} f(x)$. Thus, we have shown the necessity of existence of minimizers on compact sets.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Exercise 1.8 (Existence of Unconstrained Minimizers):

Any coercive closed function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ has a minimizer.

Proof:

Let f be a coercive and closed function. Choose $r \in \mathbb{R}$ such that the sublevel set

$$\mathcal{L} := \{x \in \mathbb{E} : f(x) \leq r\}$$

is nonempty. Because f is closed, we know that $\mathcal{L}$ is closed, and because f is coercive, it follows that $\mathcal{L}$ is compact.

By Exercise 1.6, f attains its minimum over the compact set $\mathcal{L}$, so there exists $x^* \in \mathcal{L}$ such that

$$f(x^*) = \min_{x \in \mathcal{L}} f(x).$$

Let $\alpha = \inf_{x \in \mathbb{E}} f(x)$. Since $f(x^*) \in \mathcal{L}$, it follows that $f(x^*) = \alpha$, and hence x^* is a global minimizer of f over $\mathbb{E}$.

Here, the mistake is that ChatGPT said that

Because f is closed, we know that $\mathcal{L}$ is closed.

That is not necessarily true. Rather, we should do a proof which we have done. We will reiterate it:

Proof. A function $f : \mathbb{E} \rightarrow \overline{\mathbb{R}}$ is lower semicontinuous at a point $x \in \mathbb{E}$ if we have that $\liminf_{y \rightarrow x} f(y) \geq f(x)$. Equivalently, for every $r \in \mathbb{R}$, the set $\{x \in \mathbb{E} : f(x) > r\}$ is open. Taking complements, $\{x \in \mathbb{E} : f(x) \leq r\}$ is closed. So, we have that our sublevel set of f is closed. $\square$

6. **Exercise 1.9:** Define the function $f(x) = \frac{1}{2}\langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c$, where $\mathcal{A} : \mathbb{E} \rightarrow \mathbb{E}$ is a linear operator, v lies in $\mathbb{E}$, and c is a real number.

1. Show that if $\mathcal{A}$ is replaced by the self adjoint operator $(\mathcal{A} + \mathcal{A}^*)/2$, the function values $f(x)$ remain unchanged.

Let's define a new function $g(x)$ and in $g(x)$ we replace $\mathcal{A}$ with $\frac{1}{2}(\mathcal{A} + \mathcal{A}^*)$. So, $g(x) = \frac{1}{2}\langle \frac{1}{2}(\mathcal{A} + \mathcal{A}^*)x, x \rangle + \langle v, x \rangle + c$.

So, we have that

$$\begin{aligned}
 g(x) &= \frac{1}{2} \left\langle \left(\frac{1}{2}(\mathcal{A} + \mathcal{A}^*) \right) x, x \right\rangle + \langle v, x \rangle + c \\
 &= \frac{1}{2} \left\langle \left(\frac{1}{2}\mathcal{A} + \frac{1}{2}\mathcal{A}^* \right) x, x \right\rangle + \langle v, x \rangle + c \\
 &= \frac{1}{2} \left\langle \frac{1}{2}\mathcal{A}x + \frac{1}{2}\mathcal{A}^*x, x \right\rangle + \langle v, x \rangle + c \\
 &= \frac{1}{4} \langle \mathcal{A}x + \mathcal{A}^*x, x \rangle + \langle v, x \rangle + c \\
 &= \frac{1}{4} (\langle \mathcal{A}x, x \rangle + \langle \mathcal{A}^*x, x \rangle) + \langle v, x \rangle + c \\
 &= \frac{1}{4} (\langle \mathcal{A}x, x \rangle + \langle x, \mathcal{A}x \rangle) + \langle v, x \rangle + c && \text{(By using a property of adjoints)} \\
 &= \frac{1}{4} (\langle \mathcal{A}x, x \rangle + \langle \mathcal{A}x, x \rangle) + \langle v, x \rangle + c && \text{(By using symmetry as we are working with } \mathbb{E} \text{)} \\
 &= \frac{1}{4} \langle \mathcal{A}x, x \rangle + \frac{1}{4} \langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c \\
 &= \underbrace{\frac{1}{2} \langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c}_{f(x)}
 \end{aligned}$$

We see that $g(x) = \frac{1}{2}\langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c$, which is just $f(x)$. Thus, $f(x) = g(x)$ meaning that if $\mathcal{A}$ is replaced by the self adjoint operator $(\mathcal{A} + \mathcal{A}^*)/2$, the function values $f(x)$ remain unchanged.

Extra Credit: After completing this part of the exercise, I asked ChatGPT to do the question and this is what it stated:

Let us again define the function:

$$f(x) = \frac{1}{2} \langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c$$

We claim that replacing $\mathcal{A}$ with $\frac{1}{2}(\mathcal{A} + \mathcal{A}^*)$ does **not** preserve the function value.

Let $\mathcal{S} := \frac{1}{2}(\mathcal{A} + \mathcal{A}^*)$. Then,

$$f_{\text{new}}(x) = \frac{1}{2} \langle \mathcal{S}x, x \rangle + \langle v, x \rangle + c$$

Now, notice that:

$$\langle \mathcal{S}x, x \rangle = \left\langle \frac{1}{2}(\mathcal{A} + \mathcal{A}^*)x, x \right\rangle = \frac{1}{2} \langle \mathcal{A}x, x \rangle + \frac{1}{2} \langle \mathcal{A}^*x, x \rangle$$

Since $\mathcal{A}^*x = \mathcal{A}x$ for any operator $\mathcal{A}$, we get:

$$\langle \mathcal{S}x, x \rangle = \frac{1}{2} \langle \mathcal{A}x, x \rangle + \frac{1}{2} \langle \mathcal{A}x, x \rangle = \langle \mathcal{A}x, x \rangle$$

Therefore:

$$f_{\text{new}}(x) = \frac{1}{2} \langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c = f(x)$$

Thus, even though we replaced $\mathcal{A}$ with a different operator, the function value is still the same.

The mistake here is that in general it is not the case that $\mathcal{A}^*x = \mathcal{A}x$ for any operator $\mathcal{A}$, which ChatGPT stated. Here, ChatGPT is wrong. Rather, we can say that $\mathcal{A}^*x = \mathcal{A}x$ here because $\mathcal{A}$ serves as our self adjoint. Something that satisfies our adjoint is that for a linear map $t : \mathbb{E} \rightarrow \mathbb{Y}$, there exists a unique linear map $t^* : \mathbb{Y} \rightarrow \mathbb{E}$, called the adjoint, satisfying $\langle tx, y \rangle_{\mathbb{Y}} = \langle x, t^*y \rangle_{\mathbb{E}} \quad \forall x \in \mathbb{E}, \forall y \in \mathbb{Y}$. Furthermore, a linear map $t : \mathbb{E} \rightarrow \mathbb{E}$ is called a self adjoint if $t^* = t$, which we use and apply.

2. Assuming $\mathcal{A}$ is self adjoint, derive the equations:

$$\nabla f(x) = \mathcal{A}x + v \quad \text{and} \quad \nabla^2 f(x) = \mathcal{A}$$

$$\nabla f(x) = \nabla \left(\frac{1}{2} \langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c \right)$$

$$\nabla f(x) = \nabla \left(\frac{1}{2} \langle \mathcal{A}x, x \rangle \right) + \nabla \langle v, x \rangle + \nabla c$$

$$\nabla f(x) = \frac{1}{2} \nabla \langle \mathcal{A}x, x \rangle + v$$

$$\nabla f(x) = \frac{1}{2} (2\mathcal{A}x) + v$$

$$\nabla f(x) = \mathcal{A}x + v$$

$$\nabla^2 f(x) = \mathcal{A}$$

Note that we can state that $\nabla \langle \mathcal{A}x, x \rangle = 2\mathcal{A}x$ as $\mathcal{A}$ is self adjoint. Thus, we have shown that $\nabla f(x) = \mathcal{A}x + v$ and $\nabla^2 f(x) = \mathcal{A}$.

Extra Credit: After completing this part of the exercise, I asked ChatGPT to derive the questions and this is what it stated:

Let $f(x) = \frac{1}{2} \langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c$,

where $\mathcal{A} : \mathbb{E} \rightarrow \mathbb{E}$ is a **self-adjoint** linear operator, $v \in \mathbb{E}$, and $c \in \mathbb{R}$.

Then we compute the gradient by treating the first term as a quadratic form and applying linearity:

$$\nabla f(x) = \frac{1}{2} \nabla \langle \mathcal{A}x, x \rangle + \nabla \langle v, x \rangle + \nabla c$$

Now:

- $\nabla \langle \mathcal{A}x, x \rangle = 2\mathcal{A}x$
- $\nabla \langle v, x \rangle = v$
- $\nabla c = 0$

So:

$$\nabla f(x) = \frac{1}{2} (2\mathcal{A}x) + v = \mathcal{A}x + v$$

and for the Hessian:

$$\nabla^2 f(x) = \nabla^2 \left(\frac{1}{2} \langle \mathcal{A}x, x \rangle \right) = \mathcal{A}$$

Here, the error made is that ChatGPT stated that $\nabla f(x) = \mathcal{A} + v$, which is wrong as it forgot the x term. Rather, $\nabla f(x) = \mathcal{A}x + v$. Note that here in the correct version we don't forget the x term.

3. Assuming $\mathcal{A}$ is self adjoint, show that f is coercive if and only if $\mathcal{A}$ is positive definite.

We will do this Proof by Biconditional:

Showing that if $\mathcal{A}$ is positive definite, then f is coercive:

Proof. We will start by having $\mathcal{A} \succ 0$ while also keeping in mind that we have $\mathcal{A}$ as self adjoint. Since $\mathcal{A} \succ 0$, then we have that $\langle \mathcal{A}x, x \rangle = R_{\mathcal{A}}(x) \|x\|^2$ by applying Rayleigh-Ritz Theorem. From Rayleigh-Ritz, we have that $R_{\mathcal{A}}(x) \|x\|^2 \geq \lambda_{\min} \|x\|^2$ for all $x \in \mathbb{E}$. Thus, $\langle \mathcal{A}x, x \rangle \geq \lambda_{\min} \|x\|^2$. Now, we have that $f(x) = \frac{1}{2} \langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c$. Applying that $\langle \mathcal{A}x, x \rangle = R_{\mathcal{A}}(x) \cdot \|x\|^2$, we have that $\frac{1}{2} \langle \mathcal{A}x, x \rangle + \langle v, x \rangle + c = \frac{1}{2} R_{\mathcal{A}}(x) \|x\|^2 + \langle v, x \rangle + c$. Thus, we can state that $f(x) = \frac{1}{2} R_{\mathcal{A}}(x) \|x\|^2 + \langle v, x \rangle + c$. Applying Cauchy-Schwarz Inequality, we get: $\langle v, x \rangle \geq -\|v\| \cdot \|x\|$. So by utilizing that $\langle \mathcal{A}x, x \rangle \geq \lambda_{\min} \|x\|^2$, $f(x) \geq \frac{1}{2} \lambda_{\min} \|x\|^2 - \|v\| \cdot \|x\| + c$. As $\|x\| \rightarrow \infty$, then $\frac{1}{2} \lambda_{\min} \|x\|^2 - \|v\| \cdot \|x\| + c \rightarrow \infty$ due to the quadratic term, $\|x\|^2$, dominating, meaning that $f(x) \rightarrow \infty$ as well for x_i as $i \rightarrow \infty$. Thus, we have that f is coercive. $\square$

Showing that if f is coercive, then $\mathcal{A}$ is positive definite:

Proof.

1. We start by having f as coercive. So, $f(x_i) \rightarrow \infty$. We want to derive a contradiction if $\mathcal{A}$ has an eigenvalue of $\lambda \leq 0$ because if it is not the case that $\mathcal{A}$ has an eigenvalue $\lambda \leq 0$, that would mean that all of the eigenvalues of $\mathcal{A}$ are greater than 0 giving us a positive definite $\mathcal{A}$ by definition.
2. Suppose for contradiction purposes that $\mathcal{A}$ has an eigenvalue $\lambda \leq 0$. If this is the case, then there would exist a unit eigenvector $u \in \mathbb{E}$ such that $\mathcal{A}u = \lambda u$ with $\lambda \leq 0$ and $\|u\| = 1$.
3. We now define a path in the direction of u . Let $x = tu$, where $t \in \mathbb{R}$ and $|t| \rightarrow \infty$. Then, we have:

$$\begin{aligned}
 f(x) &= f(tu) && (\text{As } x = tu) \\
 &= \frac{1}{2} \langle \mathcal{A}(tu), tu \rangle + \langle v, tu \rangle + c && (\text{Applying the definition of } f \text{ to } tu) \\
 &= \frac{1}{2} t^2 \langle \mathcal{A}u, u \rangle + t \langle v, u \rangle + c \\
 &= \frac{1}{2} t^2 \lambda + t \langle v, u \rangle + c && (\text{Since } \mathcal{A}u = \lambda u \text{ and } \|u\| = 1)
 \end{aligned}$$

4. Now, we analyze the Asymptotic Behavior:
 - a. If $\lambda < 0$, then the $\frac{1}{2}t^2\lambda$ term dominates and $f(tu) \rightarrow -\infty$ as $|t| \rightarrow \infty$ contradicting what it means for a function to be coercive.
 - b. If $\lambda = 0$, then $f(tu) = t\langle v, u \rangle + c$. This is linear in t , so it does not necessarily go to ∞ . We need $\langle v, u \rangle > 0$ and $t \rightarrow \infty$, but even that does not account for what happens when our $-t \rightarrow \infty$. We have to consider that the definition of a coercive function requires $f(x) \rightarrow \infty$ for all directions with $\|x\| \rightarrow \infty$. Thus, we have that if $\lambda = 0$, we fail to meet the requirements of f being coercive.

Thus, if $\lambda \leq 0$, we lead to a contradiction. This means $\lambda > 0$ for $\mathcal{A}$.

5. Conclusion: Thus, if f is coercive, we must have that $\mathcal{A}$ should have all positive eigenvalues as both zero eigenvalues and negative eigenvalues fail to uphold the requirements of f being coercive. By definition, because $\mathcal{A}$ has all positive eigenvalues, we must have that $\mathcal{A}$ is positive definite.

□

Thus, by showing that if $\mathcal{A}$ is positive definite, then f is coercive and showing that if f is coercive, then $\mathcal{A}$ is positive definite, we have shown f is coercive if and only if $\mathcal{A}$ is positive definite assuming $\mathcal{A}$ is self adjoint.

Extra Credit: After completing this part of the exercise, I asked ChatGPT to do the question and this is what it stated:

We claim: If $\mathcal{A}$ is positive semidefinite, i.e., $\langle \mathcal{A}x, x \rangle \geq 0$ for all x , then f is coercive.

Proof:

Since $\mathcal{A} \succeq 0$, we have

$$\langle \mathcal{A}x, x \rangle \geq 0 \quad \forall x$$

So the quadratic term $\frac{1}{2} \langle \mathcal{A}x, x \rangle$ is always nonnegative and increases with $\|x\|$.

Meanwhile, $\langle v, x \rangle$ is linear, and hence grows more slowly than the quadratic term.

The mistake here is that $\langle v, x \rangle$ does not grow or necessarily grow the same way or in the same direction as our quadratic term does the way ChatGPT is assuming it does. In fact, by applying Cauchy-Schwarz Inequality, we get: $\langle v, x \rangle \geq -\|v\| \cdot \|x\|$ meaning that $\langle v, x \rangle$ can be positive, negative, or zero.

7. **Exercise 1.10:** Define the function $f(x) = \frac{1}{2}\|F(x)\|^2$, where $F : \mathbb{E} \rightarrow \mathbb{Y}$ is a C^1 smooth mapping. Prove the identity $\nabla f(x) = \nabla F(x)^* F(x)$.

$$f(x) = \frac{1}{2}\|F(x)\|^2 \quad (\text{Given})$$

$$f(x) = \frac{1}{2}\langle F(x), F(x) \rangle \quad (\text{Writing the Given Equation equivalently})$$

$$\text{Let } h \in \mathbb{E}. \text{ Then, } f(x+h) = \frac{1}{2}\|F(x+h)\|^2$$

$$\text{Alternatively, } f(x+h) = \frac{1}{2}\langle F(x+h), F(x+h) \rangle$$

Since F is a C^1 smooth mapping, we have that:

$$F(x+h) = F(x) + \nabla F(x)h + o(\|h\|)$$

$$f(x+h) = \frac{1}{2}\langle F(x) + \nabla F(x)h + o(\|h\|), F(x) + \nabla F(x)h + o(\|h\|) \rangle$$

$$f(x+h) = \frac{1}{2}(\langle F(x), F(x) \rangle + \langle F(x), \nabla F(x)h \rangle + \langle \nabla F(x)h, F(x) \rangle + \langle \nabla F(x)h, \nabla F(x)h \rangle + o(\|h\|))$$

By symmetry of the inner product, $\langle F(x), \nabla F(x)h \rangle = \langle \nabla F(x)h, F(x) \rangle$.

$$f(x+h) = \frac{1}{2}(\langle F(x), F(x) \rangle + 2\langle F(x), \nabla F(x)h \rangle + \langle \nabla F(x)h, \nabla F(x)h \rangle + o(\|h\|))$$

$$f(x+h) = \frac{1}{2}(\|F(x)\|^2 + 2\langle F(x), \nabla F(x)h \rangle + \|\nabla F(x)h\|^2 + o(\|h\|))$$

$$f(x+h) = \frac{1}{2}\|F(x)\|^2 + \langle F(x), \nabla F(x)h \rangle + \frac{1}{2}\|\nabla F(x)h\|^2 + \frac{1}{2}o(\|h\|)$$

Scalars don't matter for our $o(\|h\|)$ term. So, we can rewrite:

$$f(x+h) = \frac{1}{2}\|F(x)\|^2 + \langle F(x), \nabla F(x)h \rangle + \frac{1}{2}\|\nabla F(x)h\|^2 + o(\|h\|)$$

By definition of the gradient,

$$\langle \nabla f(x), h \rangle = \langle F(x), \nabla F(x)h \rangle$$

Remember that:

$$\langle F(x), \nabla F(x)h \rangle = \langle \nabla F(x)^* F(x), h \rangle$$

So:

$$\langle \nabla f(x), h \rangle = \langle \nabla F(x)^* F(x), h \rangle$$

Remember, that h is our increment; $h \in \mathbb{E}$.

Now, we can use this fact about h in order to state $\nabla f(x) = \nabla F(x)^* F(x)$ as desired.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

We define the function:

$$f(x) = \frac{1}{2} \|F(x)\|^2 = \frac{1}{2} \langle F(x), F(x) \rangle$$

To compute the gradient, apply the chain rule:

$$\nabla f(x) = \frac{1}{2} \cdot \nabla \langle F(x), F(x) \rangle$$

Now take the derivative:

$$\nabla \langle F(x), F(x) \rangle = \nabla F(x)^* \cdot \nabla F(x)$$

So:

$$\nabla f(x) = \frac{1}{2} \cdot \nabla F(x)^* \cdot \nabla F(x)$$

The error is, here, ChatGPT uses the Chain Rule, which is incorrect; this is a mistake. Rather, we should use the definitions defined for inner products on $\mathbb{E}$ and $\mathbb{Y}$ or directional derivatives.

8. **Exercise 1.11:** Consider a function $f : \mathbb{E} \rightarrow \mathbb{R}$ and a linear mapping $\mathcal{A} : \mathbb{Y} \rightarrow \mathbb{E}$ and define the composition $h(x) = f(\mathcal{A}x)$.

1. Show that f is differentiable at $\mathcal{A}x$, then $\nabla h(x) = \mathcal{A}^* \nabla f(\mathcal{A}x)$.
 Let $h(x) = f(\mathcal{A}x)$. To compute $\nabla h(x)$, we use the definition of the gradient.
 Let $z \in \mathbb{Y}$, then we have that:

$$\langle \nabla h(x), z \rangle = \lim_{t \rightarrow 0} \frac{f(\mathcal{A}(x + tz)) - f(\mathcal{A}x)}{\|t\|}$$

Because $\mathcal{A}$ is linear, we can state:

$$\lim_{t \rightarrow 0} \frac{f(\mathcal{A}(x + tz)) - f(\mathcal{A}x)}{\|t\|} = \lim_{t \rightarrow 0} \frac{f(\mathcal{A}x + \mathcal{A}tz) - f(\mathcal{A}x)}{\|t\|}$$

And we can state this as:

$$\lim_{t \rightarrow 0} \frac{f(\mathcal{A}x + \mathcal{A}tz) - f(\mathcal{A}x)}{\|t\|} = \langle \nabla f(\mathcal{A}x), \mathcal{A}z \rangle$$

Now applying what we know with adjoint operators, we can state $\langle \nabla f(\mathcal{A}x), \mathcal{A}z \rangle = \langle \mathcal{A}^* \nabla f(\mathcal{A}x), z \rangle$.
 Since this holds for all $z \in \mathbb{Y}$, we conclude $\nabla h(x) = \mathcal{A}^* \nabla f(\mathcal{A}x)$.

2. Show that if f is twice differentiable at $\mathcal{A}x$, then $\nabla^2 h(x) = \mathcal{A}^* \nabla^2 f(\mathcal{A}x) \mathcal{A}$.
 We already know $\nabla h(x) = \mathcal{A}^* \nabla f(\mathcal{A}x)$.
 Let $h(x) = f(\mathcal{A}x)$. To compute $\nabla^2 h(x)$, we use the definition of the gradient.
 Let $z \in \mathbb{Y}$, then we have that:

$$\langle \nabla^2 h(x), z \rangle = \lim_{t \rightarrow 0} \frac{\nabla h(x + tz) - \nabla h(x)}{\|t\|}$$

Plug in $\nabla h(x) = \mathcal{A}^* \nabla f(\mathcal{A}x)$:

$$\lim_{t \rightarrow 0} \frac{\nabla h(x + tz) - \nabla h(x)}{\|t\|} = \lim_{t \rightarrow 0} \frac{\mathcal{A}^* \nabla f(\mathcal{A}(x + tz)) - \mathcal{A}^* \nabla f(\mathcal{A}x)}{\|t\|}$$

Because $\mathcal{A}$ is linear, that means $\mathcal{A}^*$ is linear.

Thus, we can factor out $\mathcal{A}^*$. That means:

$$\begin{aligned} \lim_{t \rightarrow 0} \frac{\mathcal{A}^* \nabla f(\mathcal{A}(x + tz)) - \mathcal{A}^* \nabla f(\mathcal{A}x)}{\|t\|} &= \mathcal{A}^* \left(\lim_{t \rightarrow 0} \frac{\nabla f(\mathcal{A}x + t\mathcal{A}z) - \nabla f(\mathcal{A}x)}{\|t\|} \right) \\ \mathcal{A}^* \left(\lim_{t \rightarrow 0} \frac{\nabla f(\mathcal{A}x + t\mathcal{A}z) - \nabla f(\mathcal{A}x)}{\|t\|} \right) &= \mathcal{A}^* \nabla^2 f(\mathcal{A}x) \mathcal{A}z \end{aligned}$$

So, we have that:

$$\nabla^2 h(x)z = \mathcal{A}^* \nabla^2 f(\mathcal{A}x) \mathcal{A}z$$

Since z was chosen arbitrarily with $z \in \mathbb{Y}$, we have

$$\nabla^2 h(x) = \mathcal{A}^* \nabla^2 f(\mathcal{A}x) \mathcal{A}$$

as desired.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

We define the function:

$$f(x) = \frac{1}{2} \|F(x)\|^2 = \frac{1}{2} \langle F(x), F(x) \rangle$$

To compute the gradient, apply the chain rule:

$$\nabla f(x) = \frac{1}{2} \cdot \nabla \langle F(x), F(x) \rangle$$

Now take the derivative:

$$\nabla \langle F(x), F(x) \rangle = \nabla F(x)^* \cdot \nabla F(x)$$

So:

$$\nabla f(x) = \frac{1}{2} \cdot \nabla F(x)^* \cdot \nabla F(x)$$

The mistake, here, is that ChatGPT interprets $\mathbb{Y}$ to be a probability space when it is actually supposed to represent a Euclidean space.

9. **Exercise 1.12:** Define the two sets:

$$\mathbb{R}_{++}^n = \{x \in \mathbb{R}^n : x_i > 0 \text{ for all } i = 1, \dots, n\}$$

$$\mathbb{S}_{++}^n = \{X \in \mathbb{S}^n : X \succ 0\}$$

Consider the two functions $f : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$ and $F : \mathbb{S}_{++}^n \rightarrow \mathbb{R}$ given by

$$f(x) = -\sum_{i=1}^n \log x_i \quad \text{and} \quad F(X) = -\log \det(X) \quad \text{respectively}$$

Note from basic properties of the determinant, the equality $F(X) = f(\lambda(X))$, where we set $\lambda(X) = (\lambda_1(X), \dots, \lambda_n(X))$.

1. Find the derivatives $\nabla f(x)$ and $\nabla^2 f(x)$ for $x \in \mathbb{R}_{++}^n$.

$$\nabla f(x) = \left(-\frac{1}{x_1}, \dots, -\frac{1}{x_n} \right)$$

$$\nabla^2 f(x) = \text{diag} \left(\frac{1}{x_1^2}, \dots, \frac{1}{x_n^2} \right)$$

2. Using the property $\text{tr}(AB) = \text{tr}(BA)$, prove $\nabla F(X) = -X^{-1}$ and $\nabla^2 F(X)[V] = X^{-1}VX^{-1}$ for any $X \succ 0$.

Hint: To compute $\nabla F(X)$, justify

$$F(X + tV) - F(X) + t\langle X^{-1}, V \rangle = -\log \det \left(I + tX^{-\frac{1}{2}}VX^{-\frac{1}{2}} \right) + t \cdot \text{tr} \left(X^{-\frac{1}{2}}VX^{-\frac{1}{2}} \right)$$

By rewriting the expression in terms of eigenvalues of $X^{-\frac{1}{2}}VX^{-\frac{1}{2}}$, deduce that the right hand side is $o(t)$. To compute the Hessian, observe

$$(X + V)^{-1} = X^{-\frac{1}{2}} \left(I + X^{-\frac{1}{2}}VX^{-\frac{1}{2}} \right)^{-1} X^{-\frac{1}{2}}$$

and then use the expansion

$$(I + A)^{-1} = I - A + A^2 - A^3 + \dots = I - A + \mathcal{O}(\|A\|_{\text{op}}^2)$$

whenever $\|A\|_{\text{op}} < 1$.

We'll use some ideas from Numerical Linear Algebra:

$$\log \det(I + tA) = \text{tr}(tA) + o(t) \text{ for small } t$$

Let's use the perturbation $X_t = X + tV$.

Then, $F(X_t) = -\log \det(X + tV)$.

Now, factor out X :

$$\begin{aligned} F(X + tV) &= -\log \det \left(X^{\frac{1}{2}} \left(I + tX^{-\frac{1}{2}}VX^{-\frac{1}{2}} \right) X^{\frac{1}{2}} \right) \\ &= -\log \det(X) - \log \det \left(I + tX^{-\frac{1}{2}}VX^{-\frac{1}{2}} \right) \end{aligned}$$

We can apply $F(X) = -\log \det(X)$ to say:

$$F(X + tV) - F(X) = -\log \det \left(I + tX^{-\frac{1}{2}}VX^{-\frac{1}{2}} \right)$$

Then, for small t :

$$F(X + tV) - F(X) = -t \cdot \text{tr}(A) + o(t)$$

So:

$$F(X + tV) - F(X) = -t \cdot \text{tr}(X^{-1}V) + o(t)$$

That gives us:

$$\langle \nabla F(X), V \rangle = -\text{tr}(X^{-1}V)$$

This gives us $\nabla F(X) = -X^{-1}$ as desired.

Now, we consider the Hessian. First, we note that:

$$(X + V)^{-1} = X^{-\frac{1}{2}} \left(I + X^{-\frac{1}{2}} V X^{-\frac{1}{2}} \right)^{-1} X^{-\frac{1}{2}}$$

For small $\|A\|_{\text{op}} < 1$, we have the expansion:

$$\begin{aligned} (I + A)^{-1} &= I - A + A^2 - A^3 + \dots \\ &= I - A + \mathcal{O}(\|A\|^2) \end{aligned}$$

Let $A = X^{-\frac{1}{2}} V X^{-\frac{1}{2}}$.

Using $(X + V)^{-1} = X^{-\frac{1}{2}} (I + A)^{-1} X^{-\frac{1}{2}}$, we get:

$$\begin{aligned} (X + V)^{-1} &= X^{-\frac{1}{2}} (I + A + \mathcal{O}(\|A\|^2)) X^{-\frac{1}{2}} \\ &= X^{-1} - X^{-1} V X^{-1} + \mathcal{O}(\|V\|^2) \end{aligned}$$

$$\nabla F(X + V) = -(X + V)^{-1}$$

$$\nabla F(X + V) = -X^{-1} + X^{-1} V X^{-1} + \mathcal{O}(\|V\|^2)$$

$$\nabla F(X + V) - \nabla F(X) = X^{-1} V X^{-1} + \mathcal{O}(\|V\|^2)$$

Now, we can state

$$\begin{aligned} \nabla^2 F(X)[V] &= \lim_{\|V\| \rightarrow 0} \frac{\nabla F(X + V) - \nabla F(X) - \mathcal{O}(\|V\|^2)}{\|V\|} \\ &= X^{-1} V X^{-1} \end{aligned}$$

Thus, we have shown $\nabla^2 F(X)[V] = X^{-1} V X^{-1}$ as desired.

Note: I figured out later that you might not have intended to mean Big $\mathcal{O}$ Notation when you stated $O(n)$, but I used Big $\mathcal{O}$ notation and that seemed to help.

3. Show

$$\langle \nabla^2 F(X)[V], V \rangle = \left\| X^{-\frac{1}{2}} V X^{-\frac{1}{2}} \right\|_F^2$$

for any $X \succ 0$ and $V \in \mathbb{S}^n$. Deduce that the operator $\nabla^2 F(X) : \mathbb{S}^n \rightarrow \mathbb{S}^n$ is positive definite.

We already have that $\nabla^2 F(X)[V] = X^{-1} V X^{-1}$. Thus, $\langle \nabla^2 F(X)[V], V \rangle = \langle X^{-1} V X^{-1}, V \rangle$. Remember that for $\mathbb{S}_{++}^n$, we have $X \succ 0$. Also, note $X^{-1} = X^{-\frac{1}{2}} X^{-\frac{1}{2}}$. We can also state that $X^{-\frac{1}{2}}$ is self adjoint.

Thus, we get:

$$\begin{aligned} \langle \nabla^2 F(X)[V], V \rangle &= \langle X^{-\frac{1}{2}} X^{-\frac{1}{2}} V X^{-\frac{1}{2}} X^{-\frac{1}{2}}, V \rangle \\ &= \langle X^{-\frac{1}{2}} V X^{-\frac{1}{2}}, X^{-\frac{1}{2}} V X^{-\frac{1}{2}} \rangle \\ &= \|X^{-\frac{1}{2}} V X^{-\frac{1}{2}}\|_F^2 \end{aligned}$$

Thus, we get $\nabla^2 F(X)$ is positive definite as we see $\|X^{-\frac{1}{2}} V X^{-\frac{1}{2}}\|_F^2 \geq 0$. Unless $V = \mathbf{0}$, where $\mathbf{0}$ represents a matrix filled with the number 0, we see that our operator $\nabla^2 F(X)$ is positive definite.

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

Step 3: Considering $\mathbb{S}$ as a Unit Sphere

If we now consider $\mathbb{S}$ to be the unit sphere, there is no direct impact on the gradient and Hessian results derived above, as the gradient and Hessian depend only on the positive definiteness of the matrix X . The unit sphere doesn't directly affect the matrix derivative computations, so the results remain unchanged:

The problem here is that for our purposes $\mathbb{S}^n$ is defined as the set of $n \times n$ real symmetric matrices rather than how ChatGPT interpreted it as meaning the unit sphere, which leads to a mistake in doing the exercise.

10. **Exercise 1.13:** Consider a function $f : U \rightarrow \mathbb{R}$ and two points $x, y \in U$. Define the univariate function $\varphi : [0, 1] \rightarrow \mathbb{R}$ given by $\varphi(t) = f(x + t(y - x))$ and let $x_t = x + t(y - x)$ for any t .

1. Show that if f is C^1 smooth, then equality

$$\varphi'(t) = \langle \nabla f(x_t), y - x \rangle \quad \text{holds for any } t \in (0, 1)$$

Define $x_t = x + t(y - x)$, so:

$$\begin{aligned} \varphi(t) &= f(x_t) \\ \varphi(t) &= f(x + t(y - x)) \\ \varphi'(t) &= \frac{d}{dt} f(x + t(y - x)) \\ \varphi'(t) &= \left\langle \nabla f(x_t), \frac{d}{dt}(x + t(y - x)) \right\rangle \\ \varphi'(t) &= \langle \nabla f(x_t), y - x \rangle \end{aligned}$$

2. Show that if f is C^2 smooth, then equality

$$\varphi''(t) = \langle \nabla^2 f(x_t)(y - x), y - x \rangle \quad \text{holds for any } t \in (0, 1)$$

We have that:

$$\begin{aligned} \varphi'(t) &= \langle \nabla f(x_t), y - x \rangle \\ \varphi''(t) &= \frac{d}{dt} \langle \nabla f(x_t), y - x \rangle \\ \varphi''(t) &= \left\langle \frac{d}{dt} \nabla f(x_t), y - x \right\rangle \\ \frac{d}{dt} \nabla f(x_t) &= \nabla^2 f(x_t)(y - x) \\ \varphi''(t) &= \langle \nabla^2 f(x_t)(y - x), y - x \rangle \end{aligned}$$

Extra Credit: After completing the exercise, I asked ChatGPT to do the question and this is what it stated:

We are given a function $f : U \rightarrow \mathbb{R}$ and two points $x, y \in U$. Define $\varphi(t) = f(x + t(y - x))$. Let's show that:

$$\varphi'(t) = \langle \nabla f(x_t), x - y \rangle$$

Here, ChatGPT mistakenly mixes up $x - y$ and $y - x$ as it should be trying to show $\varphi'(t) = \langle \nabla f(x_t), y - x \rangle$ rather than $\varphi'(t) = \langle \nabla f(x_t), x - y \rangle$, which is a mistake.

11. **Exercise 1.16:** Consider a function $f : U \rightarrow \overline{\mathbb{R}}$ that is C^2 smooth. Show that f is β smooth if and only if the inequality $\|\nabla^2 f(x)\|_{\text{op}} \leq \beta$ holds.

We will do this Proof by Biconditional:

Showing that if f is β smooth, then the inequality $\|\nabla^2 f(x)\|_{\text{op}} \leq \beta$ holds:

Proof.

$$\|\nabla f(y) - \nabla f(x)\| \leq \beta \|y - x\| \quad \forall x, y \in U$$

$$\text{Let's define } \psi(t) = \nabla f(x + tv)$$

$$\psi'(t) = \nabla^2 f(x + tv)v$$

$$\nabla f(x + tv) - \nabla f(x) = \int_0^t \nabla^2 f(x + sv)v \, ds$$

$$\left\| \frac{1}{t} \int_0^t \nabla^2 f(x + sv)v \, ds \right\| \leq \beta$$

$$\|\nabla^2 f(x)v\| \leq \beta$$

Taking the supremum over all $\|v\| = 1$:

$$\|\nabla^2 f(x)\|_{\text{op}} \leq \beta$$

□

Showing that if the inequality $\|\nabla^2 f(x)\|_{\text{op}} \leq \beta$ holds, then f is β smooth:

Proof. Let $x, y \in U$ and define the path $\gamma(t) = x + t(y - x)$, so that $\gamma(0) = x$, $\gamma(1) = y$, and $\gamma'(t) = y - x$. By the Fundamental Theorem of Calculus for vector-valued functions, we have:

$$\nabla f(y) - \nabla f(x) = \int_0^1 \nabla^2 f(x + t(y - x))(y - x) \, dt$$

Taking the norm of both sides gives:

$$\|\nabla f(y) - \nabla f(x)\| \leq \int_0^1 \|\nabla^2 f(x + t(y - x))(y - x)\| \, dt$$

Using the assumption that $\|\nabla^2 f(z)\|_{\text{op}} \leq \beta$ for all $z \in U$, we get:

$$\|\nabla^2 f(x + t(y - x))(y - x)\| \leq \beta \|y - x\|$$

So,

$$\|\nabla f(y) - \nabla f(x)\| \leq \int_0^1 \beta \|y - x\| \, dt$$

And this can be written as:

$$\|\nabla f(y) - \nabla f(x)\| \leq \beta \|y - x\|$$

This proves that f is β smooth.

□

Thus, by showing that if f is β smooth, then the inequality $\|\nabla^2 f(x)\|_{\text{op}} \leq \beta$ holds and showing that if the inequality $\|\nabla^2 f(x)\|_{\text{op}} \leq \beta$ holds, then f is β smooth, we have shown that f is β smooth if and only if the inequality $\|\nabla^2 f(x)\|_{\text{op}} \leq \beta$ holds.

Acknowledgment: For this question I got help from Batool.

Extra Credit: After completing the proof, I asked ChatGPT to do the question and this is what it stated:

Assume $\|\nabla^2 f(x)\|_{\text{op}} \leq \beta$.

Then by definition of operator norm, for any $x, y \in U$, we have:

$$\|\nabla f(y) - \nabla f(x)\| = \|\nabla^2 f(x)(y - x)\| \leq \|\nabla^2 f(x)\|_{\text{op}} \cdot \|y - x\| \leq \beta \|y - x\|,$$

The mistake, here, is that ChatGPT states that $\|\nabla f(y) - \nabla f(x)\| = \|\nabla^2 f(x)(y - x)\|$. This is not necessarily true. Rather, we should use this relationship to help solve the exercise:

$$\nabla f(y) - \nabla f(x) = \int_0^1 \nabla^2 f(x + t(y - x))(y - x) dt$$

Acknowledgment: Although not explicitly helped on a problem in particular by Collin, I have bounced many ideas off of Collin which have helped me on this homework.